13.4.5 Alkyl Isocyanides (Carbylamines)

Nomenclature of isocyanides

They are commonly named as Alkyl isocyanides. The IUPAC system names them as alkylcarbylamines

Table : Nomenclature of alkylisocyanides

Structural formula	Common name	IUPAC name
$CH_3 — NC$	Methyl isocyanide	Methylcarbylamine
$CH_3CH_2 — NC$	Ethylisocyanide	Ethylcarbylamine
$CH_3CH_2CH_2 — NC$	Propyl isocyanide	Propylcarbylamine
$C_6H_5 — NC$	Phenyl isocynaide	Phenylcarbylamine

13.4.6 Methods of preparation of isocyanides

1. From primary amines (carbylamines reaction)

Both aromatic as well as aliphatic amines on treatment with $CHCl_3$ in the presence of KOH give carbylamines

$$\underset{\text{Methylamine}}{CH_3—NH_2} + CHCl_3 + 3KOH \longrightarrow \underset{\text{Methyl isocyanide}}{CH_3 — N \equiv C} + 3KCl + 3H_2O$$

$$\underset{\text{Aniline}}{C_6H_5—NH_2} + CH.Cl_3 + 3KOH \longrightarrow \underset{\text{Phenyl isocyanide}}{C_6H_5 — N \equiv C} + 3KCl + 3H_2O$$

2. From alkyl halides

Ethyl bromide on heating with ethanolic solution of AgCN give ethyl isocyanide as major product and ethyl cyanide as minor product.

$$\underset{\text{Ethyl bromide}}{CH_3CH_2 — Br} + AgCN \xrightarrow[\text{heat}]{C_2H_5OH} \underset{\text{Ethyl isocyanide}}{CH_3CH_2 - NC} + AgBr$$

3. From N – alkyl formamide. By reaction with $POCl_3$ in pyridine.

$$CH_3 — NH — \overset{\overset{O}{\|}}{C} — H \xrightarrow[\text{(Pyridine)}]{POCl_3} CH_3 — N \equiv C + H_2O$$

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